

Find a Real Problem

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Identify claims beyond the evidence for A problem worth solving. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the outcome”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

6. Solve the given case

Shared markers are often lost because tray labels are unclear.

State the need before selecting AI.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 29 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Choose the Classroom Sorter's real user and measurable need; keep an AI-free solution option open.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Write the problem without prescribing a tool

Use the ten-request activity log to write a problem statement with a user, setting and observed difficulty. Explain why “Build an AI app” is not yet a problem statement.

My reasoning and evidence

13. W2 Identify claims beyond the evidence

A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

My reasoning and evidence

14. W3 Choose information that serves the outcome

The paper project must suggest an activity compatible with available time and a quiet-room requirement. Choose relevant inputs from minutes available, quiet required, pupil name and home address. State the output and a simpler alternative to a recommender. Explain one boundary.

My reasoning and evidence

15. W4 Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

My reasoning and evidence

16. W5 Measure help as well as avoided errors

For W4, a proposed box-finder never selects a wrong box because it always reports no match. Explain why counting only wrong suggestions would hide a failure. Name two results to record on test requests and explain why the original three-of-eight observation is not already an evaluation of the new tool.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Find a Real Problem

A fictional class loses time finding the correct craft tray.

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Find a Real Problem

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe problem

Three mismatches are recorded out of eight requests.



2. Define need

Help users choose a tray containing every requested tool.



3. Set success test

Compare selected tray against the approved inventory for each new case.



4. Bound the project

Use fictional tool requests; do not collect identity or purchase equipment.

CHANGE ONE THING

The approved list contains no tray with every requested tool.

Explain the changed result using the matching lesson.

Write a one-sentence project goal and say what evidence would count as a failure.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.
